1. There are 25 binary variables, one for each region, marking whether a tower is installed there. The objective minimizes towers installed, subject to every region having a tower in it or in an adjacent region, using the adjacency matrix from the CSV. The optimal solution installs 7 towers, in regions 1, 3, 10, 12, 13, 21, and 24. Other sets of 7 towers also cover the city, but no set of 6 does.

2. Since there are many identical items of each type, 6,720 integer variables count how many items of each of the 15 types go in each of 448 boxes, plus 448 binary variables marking whether each box is used. The objective minimizes boxes used, subject to every item being packed and no box exceeding 50 cubic feet. The upper bound of 448 boxes came from a first-fit decreasing heuristic, which places items largest first into the first box with room. The best solution found in 10 minutes uses [X] boxes ($[X]), and the best bound is 430 boxes. Since 21,484 cubic feet needs at least 430 boxes of 50.

3. There are 1,225 binary variables, one for each possible pair of the 50 teams. The objective minimizes total travel distance, using haversine distances, subject to each team playing exactly one match. The optimal solution has 25 matches totaling 4,600.5 miles and solved in [X] seconds. Most matches pair nearby rivals, such as Stanford and San Jose State (17.1 miles), while the longest is New Mexico and UNLV (483.9 miles).

4. (a) A heuristic tour can be judged without knowing the optimal tour by comparing it to a lower bound like the minimum spanning tree. Christofides uses this idea to guarantee a tour within 1.5 times the optimal length.
(b) Ant colony optimization has many simulated ants build tours, favoring short edges with more pheromone. Shorter tours deposit more pheromone and old pheromone evaporates, so edges in good tours become more likely to be chosen over time.

(c) Using nearest neighbor, each truck leaves the depot and goes to the closest unvisited customer until it has served 5, then returns, and the next truck repeats with the remaining customers. This is feasible and keeps each truck's stops close together. The drawback is that it is greedy, so the last truck may be left with scattered customers and long trips.